Project Title

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Abstract

1 (temp) Current Progress

- Data cleaning and normalisation
  - Parsed out primary vs feature artists from titles via regex pattern matching
  - Created new 'Primary Artist' and 'Featuring Artist' features
  - Created helper datasets:
    - Unique Artist , including mean feature values
    - Unique Songs dataset
    - Temporal dataset with new primary/featuring artist features
  - Rescaled 'Loudness' Feature
  - Noticed all instances of 'na' had been removed from the data set. For example, there is an entry for 'il tech' as a song by Jack Harlow, which corresponds to the true song 'Nail tech'
- Feature Engineering/Addition
  - **Lyrics and language information**: used Genius API, and lingua-py to analyse language. Had to manually correct about 30 misclassification's
  - **Retrieved additional spotify metrics**: including track popularity, artist followers, artist popularity, artist genres, and explicit flag for tracks
  - **One-hot encoded nationality, continent, languages**: for use with PCA
  - **Genre Processing:** Created 'super-genre' categories
  - **Artist Profile generation** We built artist profiles with per metrics means and standard deviations for the 7 given audio features, for use in artist prediction
- Exploratory Data Analysis
  - **Correlation analysis of original features:** discovered no surprising correlations within data
  - **Genre Distributions:**
    - By retrieving artist genres from spotify API, we find distribution of genres over the entire dataset.
    - Due to a bug with the Spotify API, not all artists return a genre
    - Songs with no genre (due to no artist genre) = 3890
    - Songs with multiple genres = 2425
    - Songs with a single genre = 2846
  - **Lyrical Analysis:** Performed basic NLP tasks on track lyrics to foster analysis of lyrics and how they may tie in to rankings and preference
  - **PCA**: Performed initial PCA on track features, and one hot encoded nationality/continent/language/explicit. Concerned not enough features for PCA to be of use.

- Modelling/Experimentation
  - **Guess the Artist, given a song**
    - We model each artist's music as a Gaussian Distribution over their song metrics
    - We then compute how likely a given song is to be by an artist (via likelihoods)
  - **Feature-based track recommender:** using track features, built basic song recommender, given a song, which suggests similar songs. Used cosine similarity as metric
  - **Hybrid feature + lyric vs. TF-IDF recommender:** Built upon previous recommender with semantic analysis and TF-IDF. Compared TF-IDF only vs. semantic only vs. hybrid vs. features only vs. random recommendations via silhouette score (cluster overlap), avg. recommendation consistency (feature similarity between recommendations and the query), artist coherence score (clustering within artist discography), avg. pairwise distance and embedding efficiency.
  - **Genre semantic clustering:**
    - Explored both k-means and agglomerative clustering on genre text embeddings. Limited relevant results, which motivated our curation of 'super-genres'
    - *Method:* extract artist genres from track, if multiple, assign individual genres to super-class, then take most common superclass as the track superclass
    - E.g: the song dákiti by bad bunny and jhayco, has these genres attached: reggaeton, trap latino, urbano latino, latin. Reggaeton is electronic, trap latino is trap, urbano latino is dance, latin is dance. Thus, the overall genre of the song is dance.
- Finalise supplementary task

2 (temp) Plans for completion

1. Refactor and clean code base, aggregate datasets
2. Primary Task: Genre super-class prediction:
   - Finish super-genre mapping
   - Decide on models on to perform genre prediction (randomforest, log reg, non-linear methods)
   - Determine metrics: precision, F1, accuracy, means/std
   - error analysis
3. Secondary task: Track recommender
4. Consider tasks using temporal patterns and rank, or collaborations
5. Writing
   - Populate introduction, background

3 Introduction

- Introduction
  - description of the task/objective
  - relevant background and related previous work
  - explanation of the significance/relevance of the objective/task

4 Data Preparation

Description of Dataset The original dataset consists of Spotify's 'Top 200' playlists collected from across the globe, from January 2017 to May 2023. It includes 651,936 rows, each of which correspond to the appearance of that song within the 'Top 200' playlist for a given day. Entries contain metadata about the song (e.g. Title, Date, Song URL) and data about the artist (e.g. Nationality, Continent). We utilise the additional track audio features provided by the User-engagement and Metrics team (e.g. energy, danceability, acousticness) and the custom 'Points (Total)' (based on rank) to aid our exploration of the data.

Data Cleaning Initial inspection of the data highlighted several aspects of the data that required cleaning. We note: **(1)** entries were duplicated depending on the number of artists (one for each artist), differing only by artist metadata and split of assigned points. **(2)** The 'Loudness' feature appears to be improperly scaled.

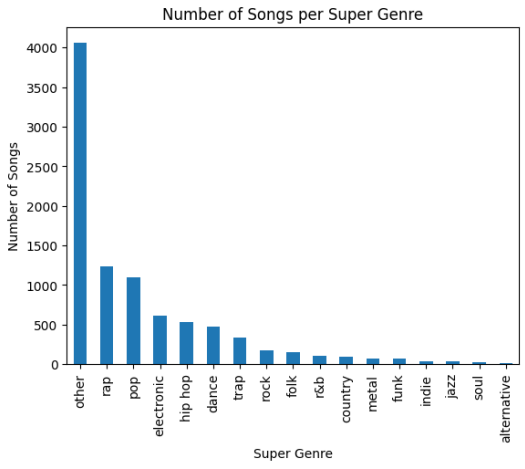
To address duplicate entries, we consolidate rows by creating 'Primary Artists' and 'Featuring Artists' columns, extracted via regex expressions on song titles, resulting in a reduced 467,070 entries. To rescale 'Loudness', we assume the values of 'Loudness' to correspond to dBFS (Decibels relative to Full Scale), which for digital audio typically fall into a range of [-22 (very quiet) to 0 (maximum loudness)]. A histogram of the original 'Loudness' values (shown in Appendix A) revealed values were heavily skewed to a range of [-20000,-2500], with a small subset appearing to reflect the likely correct dBFS values. We rescale the majority of skewed values by a factor of 1000, and manually adjust 3 entries with positive 'Loudness' values.

Multiple Datasets We divide our dataset into two key splits: one with temporal data, and one without. Each split has multiple refined datasets, depending on the task. The dataset with temporal data allows analysis over time of the data. The dataset without temporal data allows us to explore the underlying relations between songs, artists, and their features.

Feature Enrichment Next, we enrich our dataset with the following additional features from the Genius (CITE) and Spotify (CITE) APIs: `lyrics, artist_id, explicit, track_popularity, artist_followers, artist_genres, artist_popularity`. Additionally, we utilise the `lingua-py` python library to extract the language(s) of each song, adding the following features `detected_languages, num_languages, language_confidences`. Additionally, to perform PCA on our pre-processed dataset, we one-hot encode the `detected_languages, continent, nationality` features in our non-temporal dataset. This results in 131 features, for 9161 unique songs.

'Super-Genre' Mapping The original dataset has no information regarding track genres. We utilise the Spotify API to extract genres associated with the artist of each song. This resulted in 346 unique genres, from psychedelic rock, to emo rap. A limitation in the Spotify API is that not all artist's had an assigned set of genres. For example, 'lil secret' by Jack Harlow has no assigned genre, as the Spotify API call for Jack Harlow returned a blank list of genres.

We decided on the following list of 'super-genres' to map existing sub-genres into: `rap, pop, electronic, hip hop, dance, trap, rock, folk, r&b, country, metal, funk, indie, jazz, soul, alternative, other`. The 'other' super-genre contains those genres which do not easily fall into the previously defined categories. These include: `["christmas", "comedy", "ghazal", "hollands", "murga", "surf", "soundtrack", "música mexicana", "musicals", "NaN]`. Note, this includes any tracks without an assigned genre(s). We can see from Figure **??**, that a significant number of tracks within our cleaned dataset (specifically 3890) have no assigned genre. The next most popular genre is rap, then pop and so on.



5 Exploratory Data Analysis

To better understand our updated dataset, and guide the remainder of our project, we utilise several exploratory data analysis methods: Principal Component Analysis (PCA), features cor

Correlation Heatmap To gain an insight into surface level connections between our numerical features, we produce a correlation heatmap, shown in Figure 1. Certain obvious correlations appear, such as 'Valence' and 'Energy' (positive), and 'Energy' and 'Acousticness' (negative). The heatmap shows there are no surprising correlations in the data, so we further examine further exploratory data analysis methods.

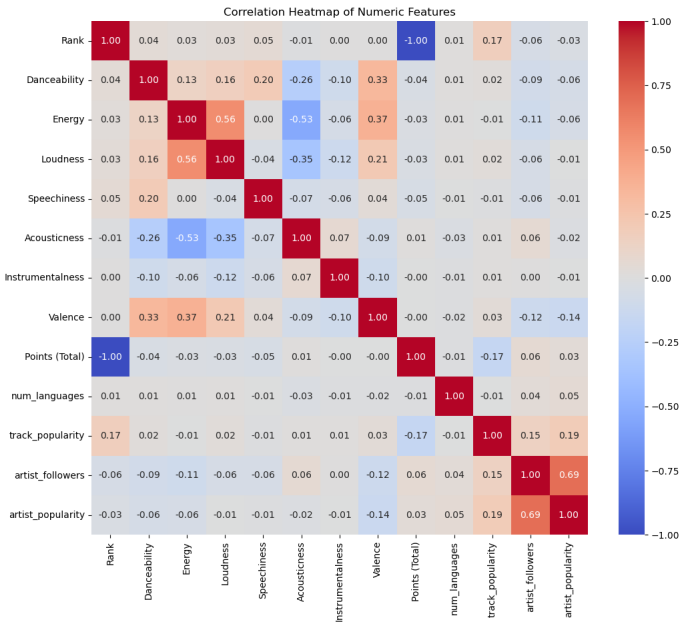


Figure 1: Cleaned Features Correlation Heatmap

Principal Component Analysis In order to glean meaningful insights from our newly enriched dataset, we perform PCA on the 7 audio features, along with one-hot encodings for nationality, continent, and language. We normalise our data prior to performing PCA. We visualise the first and second principal components in Figure

6 Primary Task: Genre Prediction

7 Supplementary Task: Song Recommendation

8 Results

9 Conclusion

Citations and References

We show a minimal example of how to use it here.

LaTeX [2] is a set of macros built atop TeX [1].

References

[1] Donald E. Knuth. *The TeX Book*. Addison-Wesley Professional, 1986.

[2] Leslie Lamport. *LaTeX: a Document Preparation System*. Addison Wesley, Massachusetts, 2 edition, 1994.

A Supplementary Figures

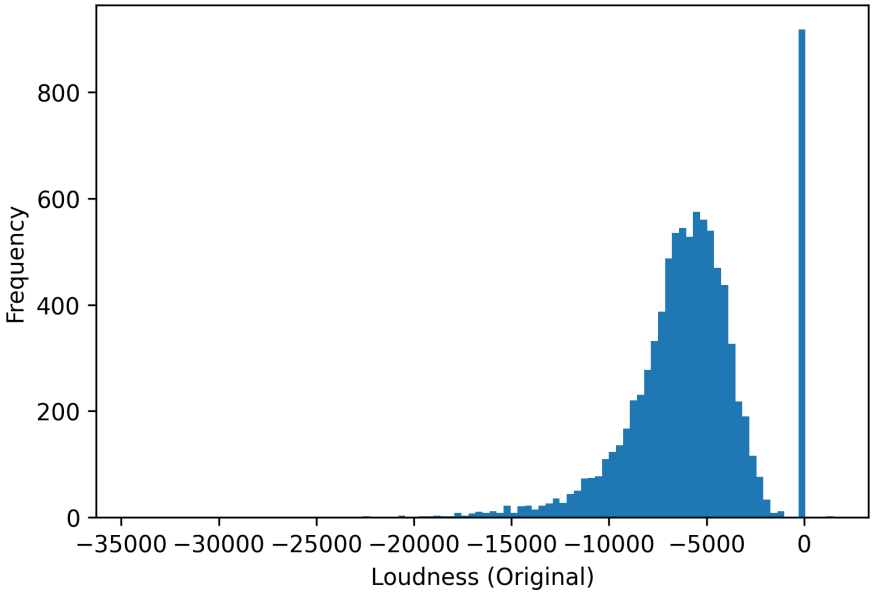


Figure 2: Original loudness distribution showing extreme outliers and irregular scaling.

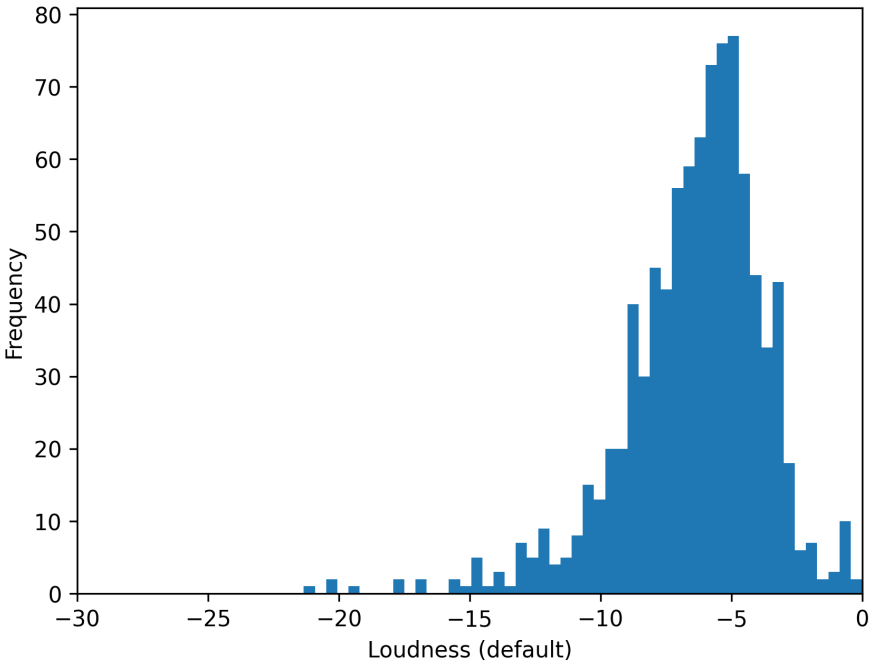


Figure 3: Distribution after clipping to realistic digital audio levels (–22 dB to –2.5 dB).

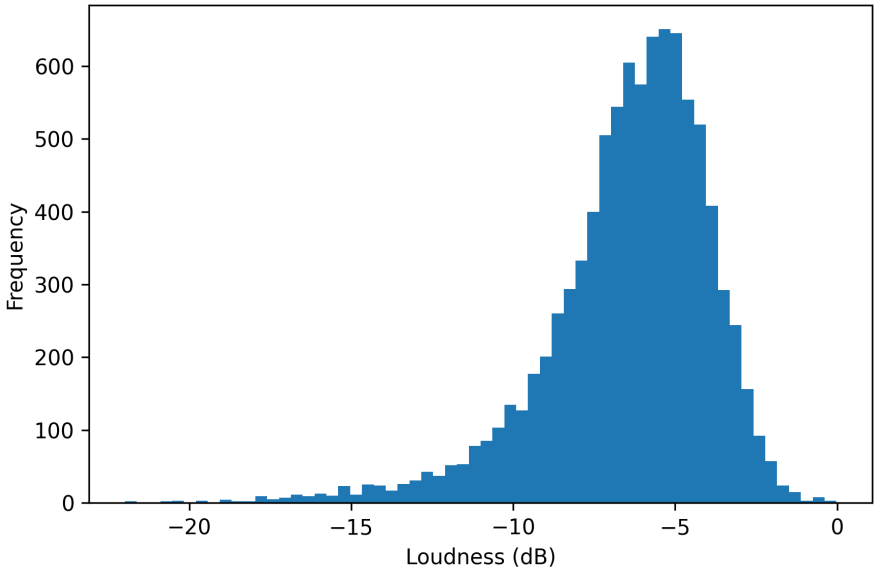


Figure 4: ‘Zoomed in’ Original Loudness values within typical range